

Special issue in memory of

H.S. Sichel



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assisted by L.P. Fatti and J.S. Galpin

Editorial note

This issue of the *South African Statistical Journal* commemorates the life and work of one of our colleagues who was respected and revered by all who knew him — **Herbert Sichel**. The accompanying obituary, which appeared in *Amstat News* (1995), pays tribute to his many contributions to Statistics and this is an indication of his standing in the international community of scholars. It stresses, in particular, “Sichel’s t ” and the Generalised Inverse Gaussian Poisson distribution to which much of his time was devoted, and mentions many of the other topics in the fields of psychology, marketing, operations research, geostatistics and bibliometrics to which he made notable contributions. But during his last years in the Department of Statistics and Actuarial Science at the University of the Witwatersrand, and especially after his final retirement, he also devoted time to studying a number of new four-parameter distributions capable of providing improved fits to obstreperous data-sets, particularly to those occurring in the mining industry.

The distribution which he thought held most promise is the logarithmic generalised inverse Gaussian distribution (LNGIG) (see Sichel, Kleingeld and Assibey-Bonsu, 1992); and, in addition to including a previously unpublished manuscript of his on the GIGP, our memorial issue contains a contribution, which is submitted jointly with Dohm and Kleingeld, on this distribution.

There are two aspects of the LNGIG which appealed to Sichel and which he hoped to exploit for practical purposes.

1. Taking its density as $\frac{1}{2sK_\gamma(b)} \exp \left\{ \gamma \frac{x - \xi}{s} - b \cosh \frac{x - \xi}{s} \right\}$

with $x = \ln z$, or

$$\frac{1}{2szK_{\gamma}(b)} \exp \left\{ \gamma \frac{\ln z - \xi}{s} - b \cosh^{\text{iv}} \frac{\ln z - \xi}{s} \right\} \text{ for } z > 0,$$

where $-\infty < \xi < \infty$, $-\infty < \gamma < \infty$, $b > 0$ and $s < 0$, and putting

$$u = \frac{\ln z - \xi}{s}, \text{ yields a standardised form } \Omega(u) = \frac{1}{2K_{\gamma}(b)} e^{\gamma u - b \cosh u}$$

for $-\infty < u < \infty$.

In this, $K_{\gamma}(b)$ is the modified Bessel function of the second kind, of order γ and with argument b . Because of the symmetry of $\cosh u$ we need only consider $\gamma \geq 0$, and b is clearly restricted to being non-negative. To consider the behaviour of the distribution as b tends to zero, let $v = u + \ln(b/2)$ so that its density becomes

$$f_V(v) = \frac{e^{\gamma v}}{2K_{\gamma}(b) \left(\frac{b}{2}\right)^{\gamma}} \exp \left(-e^v - \frac{b^2}{4} e^{-v} \right).$$

If $\gamma > 0$, as $b \rightarrow 0$, so $K_{\gamma}(b) \left(\frac{b}{2}\right)^{\gamma} \sim \frac{1}{2} \Gamma(\gamma)$ and $f_V(v) \sim \frac{e^{\gamma v - \exp v}}{\Gamma(\gamma)}$ for $-\infty < v < \infty$, resulting in v having a log-gamma distribution with cumulant generating function easily found to be

$$K(t) = \ln \frac{\Gamma(\gamma + t)}{\Gamma(\gamma)}$$

so that the cumulants satisfy $K_r \sim \frac{(-1)^r (r-1)!}{\gamma^r}$ as $\gamma \rightarrow 0$.

These provide the entry $\gamma = 0$, $-\sqrt{\beta_1} = 2$, $\beta_2 = 9$ in table 2 of Sichel, Dohm and Kleingeld (1997).

For $\gamma = 0$, however, this argument breaks down, and it is known, (Ruhkin, 1974 or Jorgenson, 1982), that if

$$\Omega(u; a) = \frac{1}{2aK_o(b)} e^{-b \cosh u/a} \text{ for } -\infty < u < \infty,$$

and if b and a decrease to zero in such a way that $-a \ln b \rightarrow \delta > 0$ then $\Omega(u; a)$ approaches the uniform distribution on $[-\delta, \delta]$.

Figure 1 is a graph of the function $e^{-b \cosh x/a}$ in the case $a = 1/6012.5$ and $b = \frac{1}{4} \times 10^{-1305}$, for which $\delta = .5000016$, so that the normalising factor needed to make this a density function is $2aK_o(b) \doteq -2a \ln b \doteq 1$.

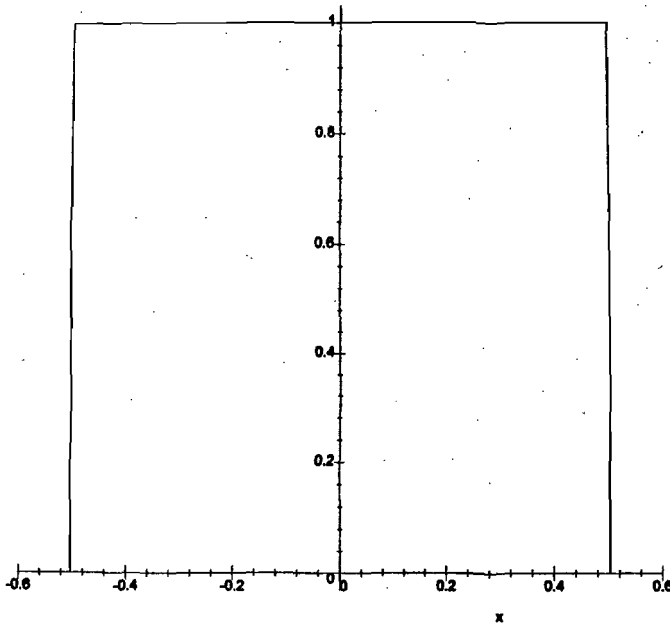


Figure 1: Density $\Omega(u; a)$ for $a = 1/6012.5$

It is clear that the corresponding density $\Omega(u; a)$ is a close approximation to the uniform distribution on $[-\frac{1}{2}, \frac{1}{2}]$, and figure 2 highlights the difference between these two, which is clearly negligible for ordinary statistical purposes. Because of this, Sichel was interested in the possibility of using the method of Sichel, Dohm and Kleingeld (1997) for fitting the LNGIG to data which are approximately uniform but with unknown end-points of the range.

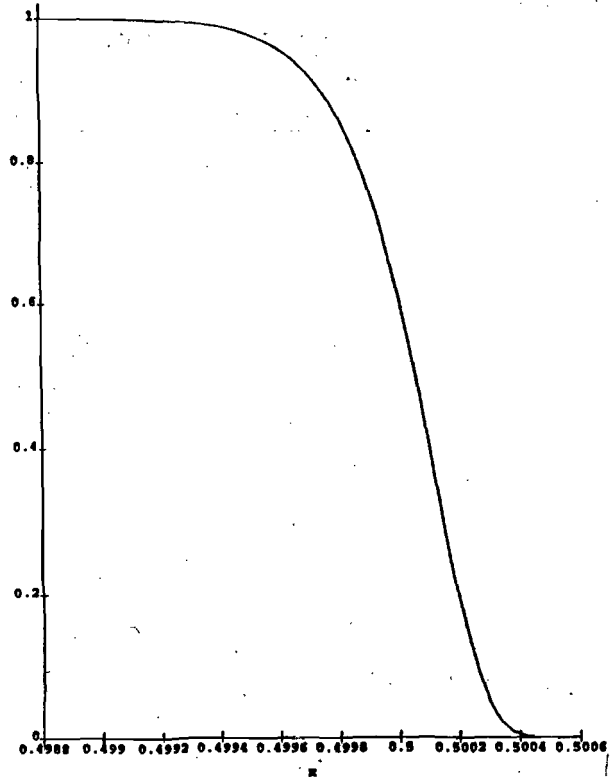


Figure 2: Difference between $\Omega(u; a)$ for $a = 1/6012.5$ and uniform distribution on $[-\frac{1}{2}, \frac{1}{2}]$

Unfortunately no record has been found of his work in this area.

It should also be remarked that, as the uniform distribution has zero skewness and kurtosis = 1.8, the corresponding point in the plot (in figure 1 of Sichel, Dohm and Kleingeld, 1997) of β_2 against $-\sqrt{\beta_1}$ does not lie on the curve relating to the log-gamma distribution. One may conjecture that as γ tends to zero the minimum turning points of the curves for fixed γ approach the point (0, 1.8) with $0 \leq -\sqrt{\beta_1} \leq 2$ and $-\sqrt{\beta_1} = 2$ with $1.8 \leq \beta_2 \leq 9$. There is a need here for further numerical investigation.

2. Another matter which Sichel considered very promising from an applied point of view was the actual existence of the non-monotonic graphs of β_2 versus $-\sqrt{\beta_1}$ for fixed γ and varying b . He regarded this as enhancing the adaptability of the LNGIG to describing data for which the standard distributions fail. As an example of the flexibility available with the LNGIG it is easily shown that it can have four points of inflection, leading to a unimodal distribution with a slight kink in one tail.

For $f(x) = \exp(\gamma x - b \cosh x)$ one has that $f''(x) = 0$ if $(\gamma - b \sinh x)^2 = b \cosh x$, from which $q(t) = (t - \gamma)^4 - (b^2 + t^2)$ is zero, where $t = b \sinh x$.

The quartic $q(t)$ has 4 distinct real roots if $q(t)$ has 3 distinct real roots; and comparison of the graphs of $(t - \gamma)^3$ and $t/2$ shows this to be the case if and only if $|\gamma| < \frac{\sqrt{6}}{9} = 0.2721655$. Hence for such γ , $f(x)$ (and so also the resulting LNGIG) has 4 points of inflection.

Figure 3 shows the graph of

$$y = \exp(-x/10 - 10^{-4} \cosh x)$$

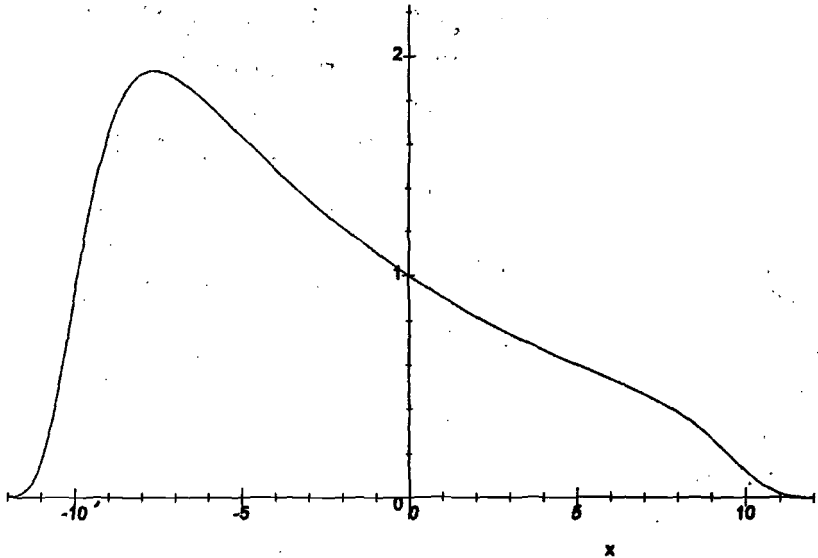


Figure 3: Graph of $y = \exp(-x/10 - 10^{-4} \cosh x)$

and suggests that the LNGIG may also be of use in describing certain mixtures of two distributions. Sichel experimented in this area with examination marks from combined classes of different levels of ability and is known to have had very encouraging results; surviving details are, however, inadequate for presentation here.

ACKNOWLEDGEMENT

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